



Intelligent Agents

Pertemuan 2 MK Kecerdasan Buatan (KOM 1327)

Jadwal Praktikum AI 2023

Kamis							
08:00 - 09:40	KOM120I - Struktur Diskret	K/3	Agus Buono	RK. A. SATARI 03.06	40	G61 (KOM)	3
08:00 - 09:40	KOM120I - Struktur Diskret	K/4	Agus Buono	RK. A. SATARI 03.07	40	G61 (KOM)	3
08:00 - 10:00	KOM1327 - Kecerdasan Buatan	P/1	Yeni Herdiyeni	Ruangan Disesuaikan	0	G61 (KOM)	5
08:00 - 10:00	KOM1327 - Kecerdasan Buatan	P/2	Yeni Herdiyeni	Ruangan Disesuaikan	0	G61 (KOM)	5
10:00 - 11:40	KOM120E - Rangkaian Digital	K/2	Auriza Rahmad Akbar	RK. A. SATARI 03.06	40	G61 (KOM)	3
10:00 - 11:40	KOM120E - Rangkaian Digital	K/4	Auriza Rahmad Akbar	RK. A. SATARI 03.07	40	G61 (KOM)	3
10:00 - 11:40	KOM133A - Sistem Informasi	K/3	Firman Ardiansyah	RK. A. SATARI 03.09	40	G61 (KOM)	5
10:00 - 12:00	KOM1314 - Komunikasi Data dan Jaringan Komputer	P/1	Heru Sukoco	Ruangan Disesuaikan	0	G61 (KOM)	5
13:00 - 14:40	KOM1303 - Analisis Algoritme	K/1	Wisnu Ananta Kusuma	RK. A. SATARI 03.11	40	G61 (KOM)	5
13:00 - 15:00	KOM120F - Basis Data	P/1	Meuthia Rachmaniah	Ruangan Disesuaikan	0	G61 (KOM)	3
13:00 - 15:00	KOM120F - Basis Data	P/3	Meuthia Rachmaniah	Ruangan Disesuaikan	0	G61 (KOM)	3
13:00 - 15:00	KOM1314 - Komunikasi Data dan Jaringan Komputer	P/2	Heru Sukoco	Ruangan Disesuaikan	0	G61 (KOM)	5
13:00 - 15:00	KOM1314 - Komunikasi Data dan Jaringan Komputer	P/3	Heru Sukoco	Ruangan Disesuaikan	0	G61 (KOM)	5
15:30 - 17:30	KOM120F - Basis Data	P/2	Meuthia Rachmaniah	Ruangan Disesuaikan	0	G61 (KOM)	3
15:30 - 17:30	KOM1327 - Kecerdasan Buatan	P/3	Yeni Herdiyeni	Ruangan Disesuaikan	0	G61 (KOM)	5

Outline

- I. Agents and environments
- II. The Concept of Rationality
- III. PEAS (Performance measure, Environment, Actuators, Sensors)
- IV. Environment types
- V. Agent types

Review



Chapter 1 identified the concept of Rational Agents.

This chapter makes this notion more concrete.

The concept of rationality can be applied to a wide variety of agents operated in any imaginable environment.

Review

The concept of rational agent is used to develop a small set of design principles for building successful agents – systems that can be called **intelligent**.



Rational agent- one that behaves as well as possible.

How well an agent can behave depends on the environment.



- I. **Agents and environments**
- II. The Concept of Rationality
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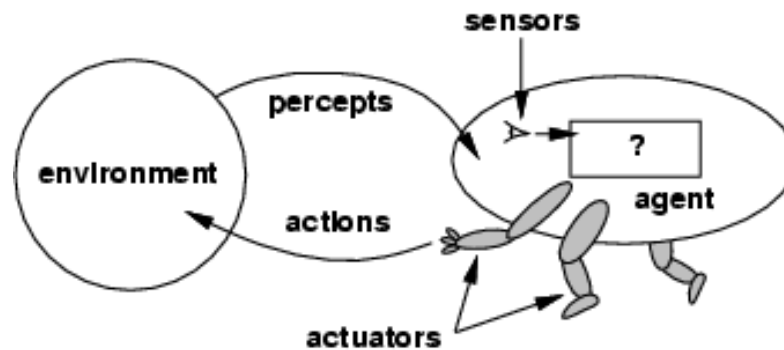
Agents

An **agent** is anything that can be viewed as **perceiving** its **environment** through **sensors** and **acting** upon that environment through **actuators**

Human agent: eyes, ears, and other organs for sensors;
hands, legs, mouth, and other body parts for actuators.

Robotic agent: cameras and infrared range finders for sensors;
various motors for actuators

Agents and environments

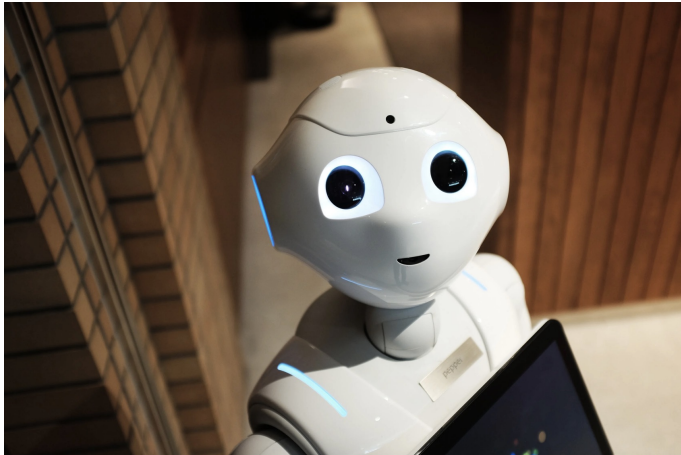


The **agent function** maps from percept histories to actions:

$$[f: \mathcal{P}^* \rightarrow \mathcal{A}]$$

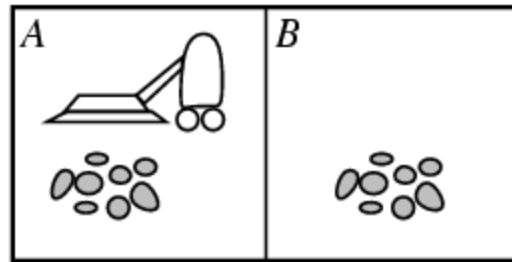
The **agent program** runs on the physical **architecture** to produce f

Examples of Intelligent Agents



An agent's choice of action at any given instant can depend on **its built-in-knowledge** and on the **entire percept sequence** to date, but not anything it hasn't perceived.

Vacuum-cleaner world



Percepts: location and contents, e.g., [A,Dirty]

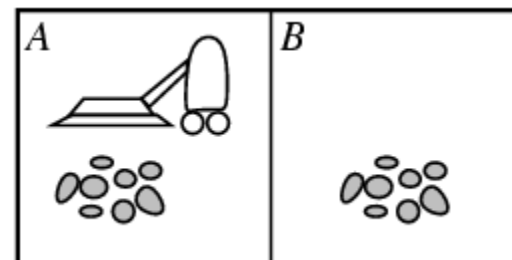
Actions: *Left*, *Right*, *Suck*, *NoOp*

Example of a very simple agent function:

if the current square is dirty, then suck, otherwise move to the other square.

A vacuum-cleaner agent

Percept sequence	Action
$[A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Dirty}]$	<i>Suck</i>
$[B, \textit{Clean}]$	<i>Left</i>
$[B, \textit{Dirty}]$	<i>Suck</i>
$[A, \textit{Clean}], [A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Clean}], [A, \textit{Dirty}]$	<i>Suck</i>
$\vdots$	$\vdots$



A Simple Agent Program

function REFLEX-VACUUM-AGENT([*location,status*]) **returns** an action

if *status* = *Dirty* **then return** *Suck*

else if *location* = *A* **then return** *Right*

else if *location* = *B* **then return** *Left*



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Rational agents

An agent should strive to "**do the right thing**", based on what it can perceive and the actions it can perform.

The right action is the one that will cause the agent to be most successful.

Therefore we will need some way to measure success.

Performance measure: An objective criterion for success of an agent's behavior

E.g., performance measure of a vacuum-cleaner agent could be amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated, etc.

Rational agents

What is rational at any given time depends on four things:

- The performance measure that defines the criterion of success.
- The agent's prior knowledge of the environment.
- The actions that the agent can perform.
- The agent's percept sequence to date.

Rational agents

Rational Agent: For each possible percept sequence, a rational agent should select an action that is expected **to maximize its performance measure**, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.



Brainy Question!

Consider the following example: I am walking along the Champs Elysées one day and I see an old friend across the street. There is no traffic nearby and I'm not otherwise engaged, so, being rational, I start to cross the street.

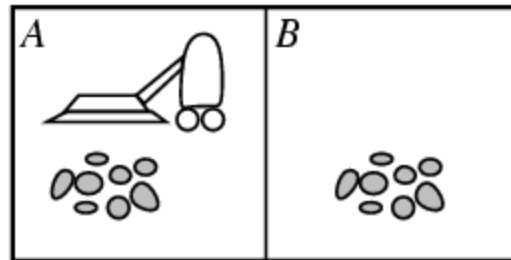
Meanwhile, at 33,000 feet, a cargo door falls off a passing airliner,³ and before I make it to the other side of the street I am flattened. Was I irrational to cross the street?

“Rational is not the same as perfection, rational maximizes expected performance. “

Going back to the Vacuum Cleaner agents

Consider the simple vacuum-cleaner agent that cleans a square if it is dirty and moves to the other square if not.

Is this a rational agent?



What is the performance measure?

What is known about the environment?

What sensors and actuators the agent has

The performance measure awards one point for each clean square at each time steps.

The “geography” of the environment is known a priori, but the dirt distribution and the initial location of the agents are not.

The only available actions are Left, Right, Suck, No Op (do nothing)

The agents correctly perceives its location and whether that location contains dirt.

The performance measure awards one point for each clean square at each time steps.

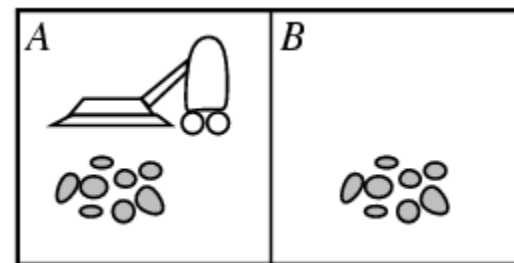


Under these circumstances the agent is **indeed rational**

The agent would be **irrational** under different circumstances.

For example once all the dirt is cleaned up it wil oscillate needlessly back and forth.

If the performance measures includes a penalty of one point for each movement left or right the agents will fare poorly.





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PEAS

PEAS: Performance measure, Environment, Actuators, Sensors

Must first specify the setting for intelligent agent design

Consider, e.g., the task of designing an automated taxi driver:

- Performance measure
- Environment
- Actuators
- Sensors



Work in group!

PEAS

Must first specify the setting for intelligent agent design

Consider, e.g., the task of designing an automated taxi driver:

- Performance measure: Safe, getting to correct destination, minimizing fuel consumption, fast, legal, comfortable trip.
- Environment: Roads, other traffic, pedestrians, customers
- Actuators: Steering wheel, accelerator, brake, signal, horn etc.
- Sensors: Cameras, sonar, speedometer, GPS etc.



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Environment types

Fully observable (vs. partially observable): An agent's sensors give it access to the complete state of the environment at each point in time.

Deterministic (vs. stochastic): The next state of the environment is completely determined by the current state and the action executed by the agent.

Episodic (vs. sequential): The agent's experience is divided into atomic "episodes" (each episode consists of the agent perceiving and then performing a single action), and the choice of action in each episode depends only on the episode itself.

Partially observable example



Environment types

Static (vs. dynamic): The environment is unchanged while an agent is deliberating.
(The environment is **semidynamic** if the environment itself does not change with the passage of time but the agent's performance score does)

Discrete (vs. continuous): A limited number of distinct, clearly defined percepts and actions.

Single agent (vs. multiagent): An agent operating by itself in an environment.

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Figure 2.6 Examples of task environments and their characteristics.

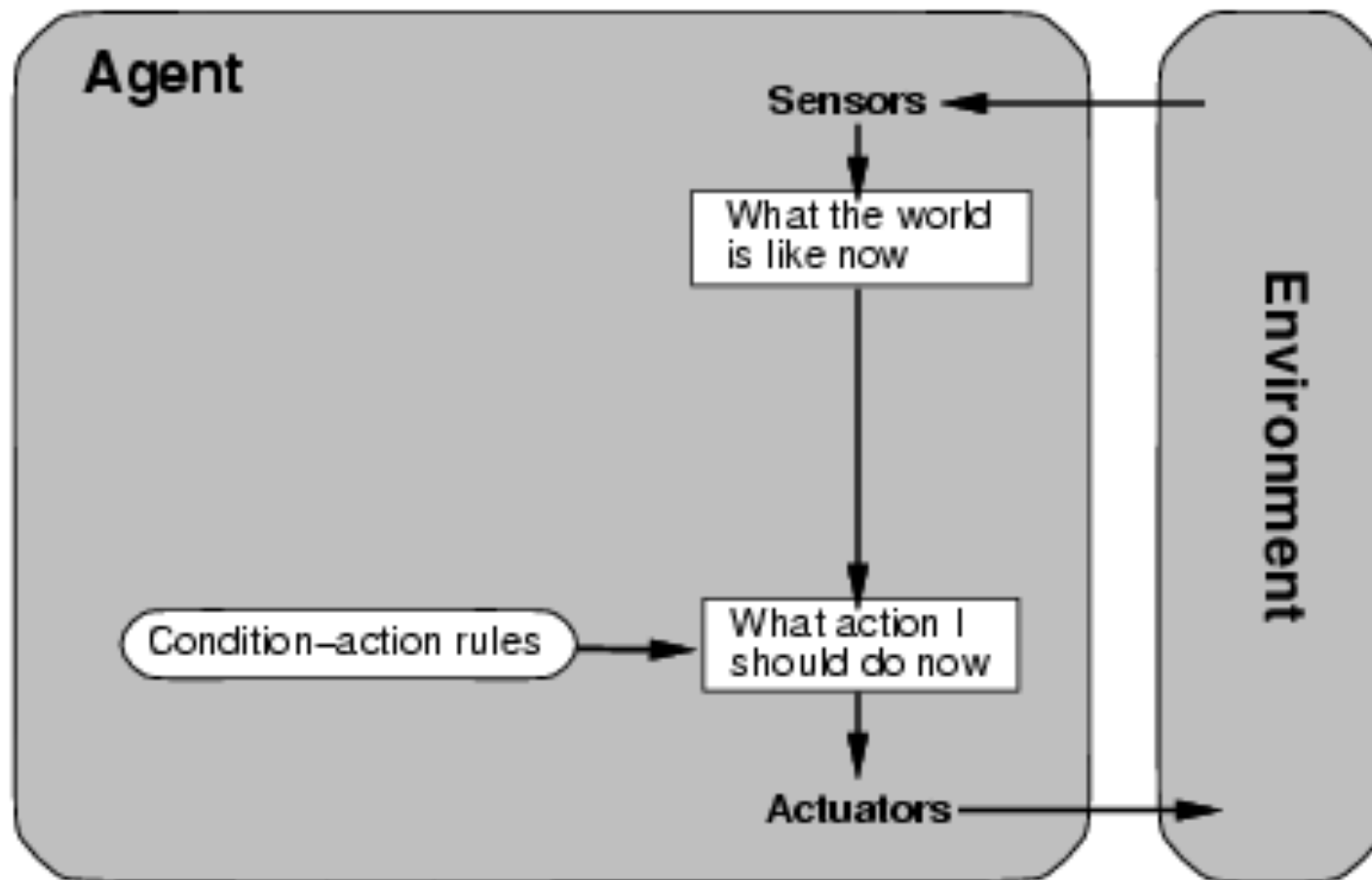


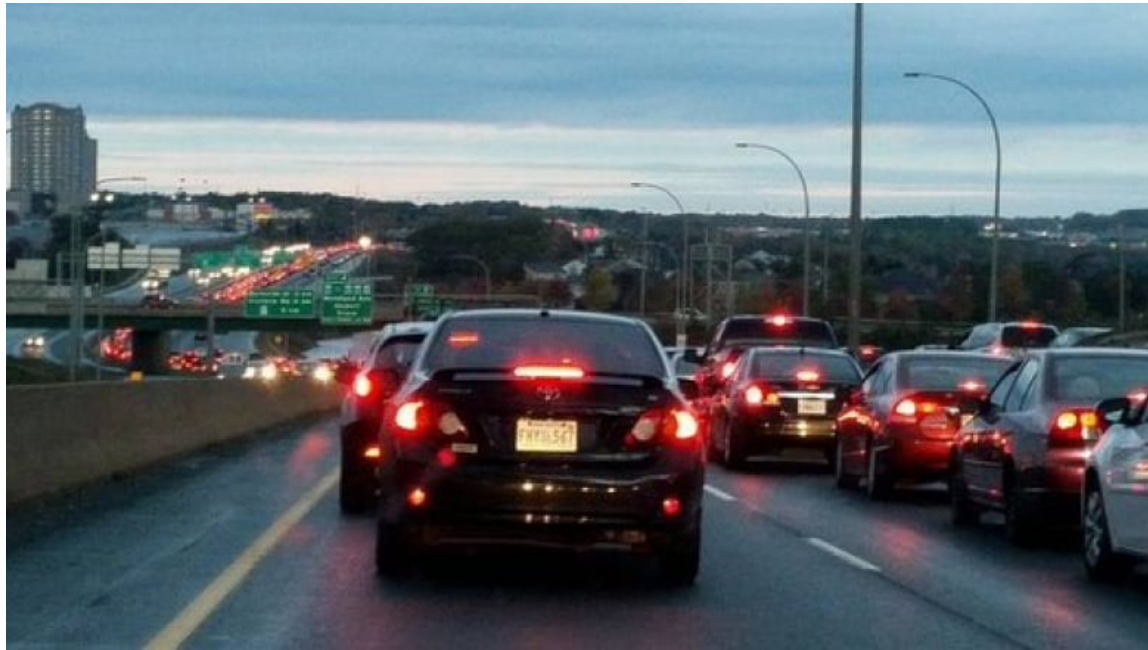
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Agent types

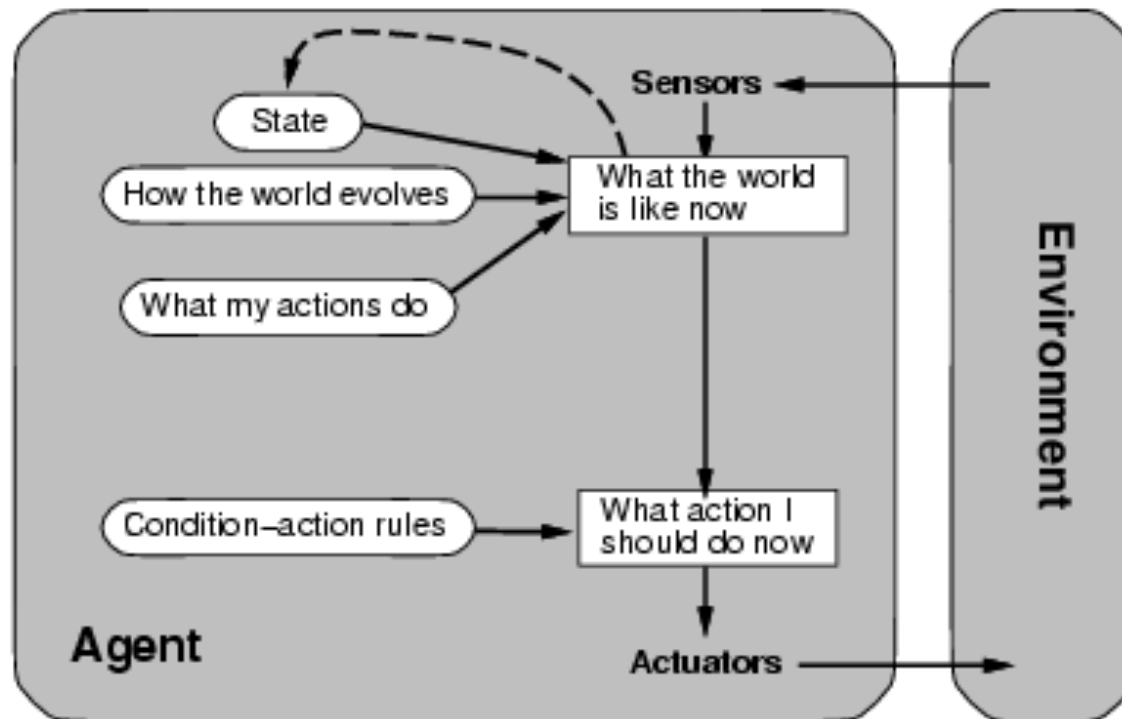
1. Simple reflex agents
2. Model-based reflex agents
3. Goal-based agents
4. Utility-based agents
5. Learning agents

Simple reflex agents

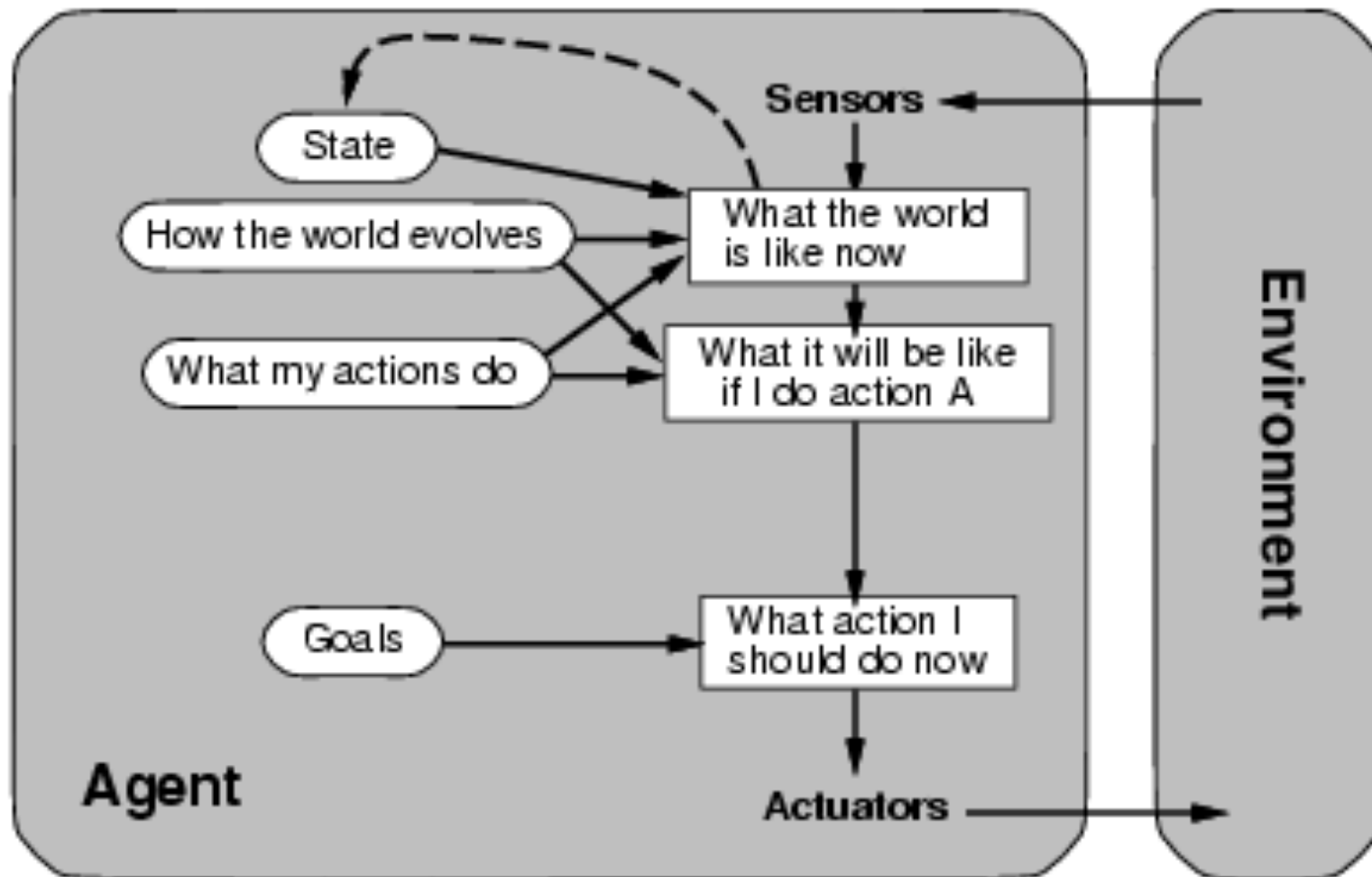




Model-based reflex agents



Goal-based agents

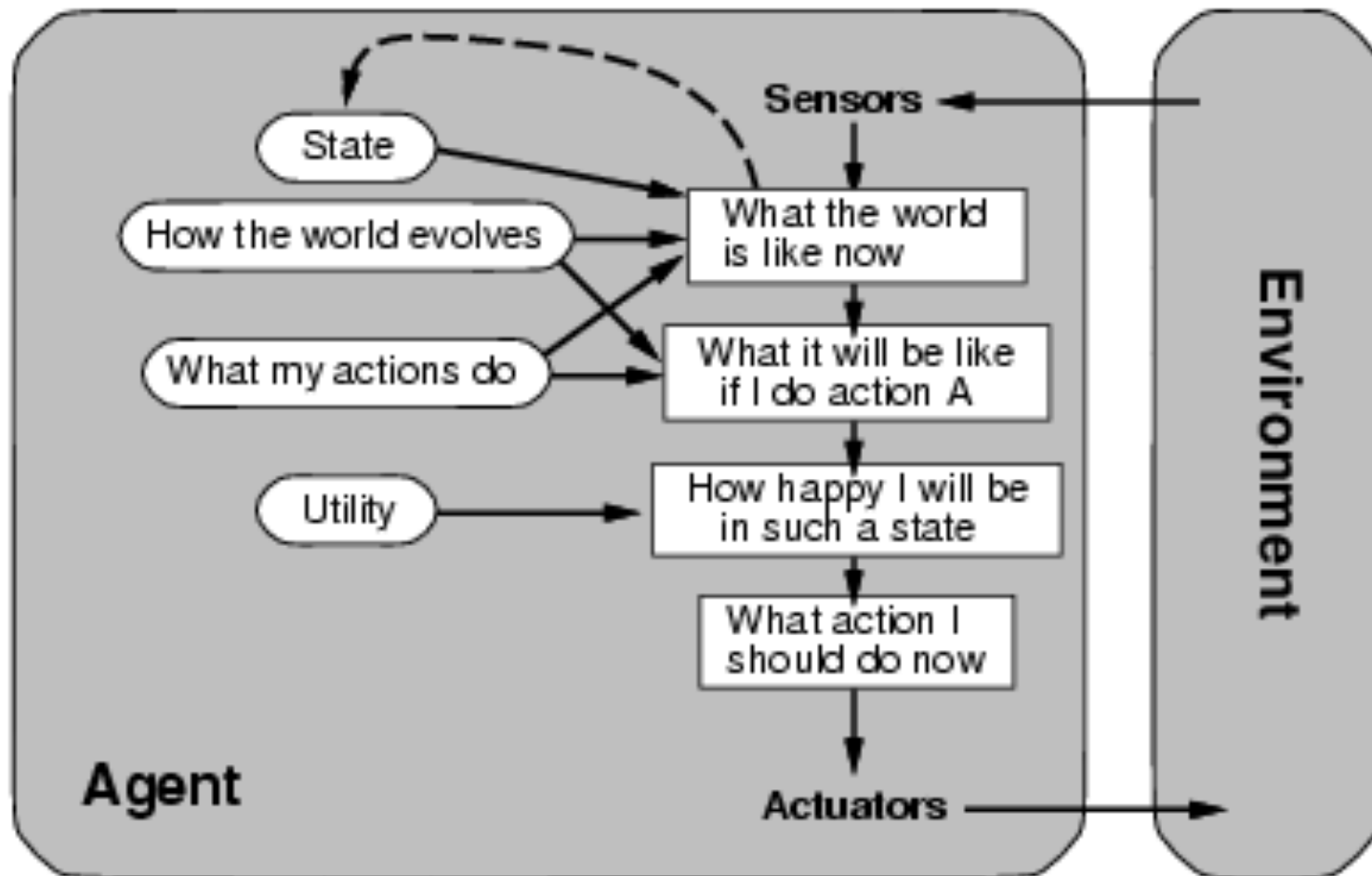


Goals just provide a crude binary distinction between "happy" and "unhappy" states.

A more general performance measure should allow a comparison of different world states according to exactly how happy they would make the agent.

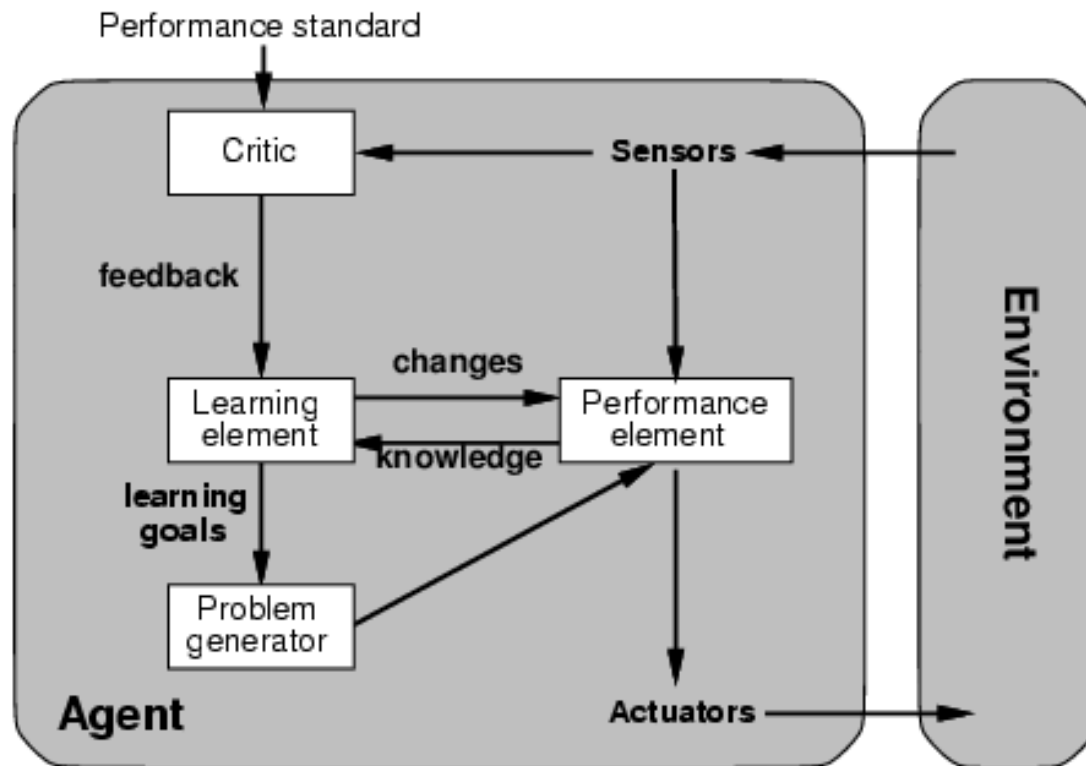
Because "happy" does not sound very scientific, economists and computer scientists use the term **utility** instead.

Utility-based agents



Technically speaking, a rational utility-based agent chooses the action that maximizes the **expected utility of the action outcomes**

Learning agents



References

Russell, S. J., Norvig, P., & Davis, E. (2010). **Artificial intelligence: a modern approach**. 3rd ed. Upper Saddle River, NJ: Prentice Hall

and other various sources.